THE DEGREE OF A LOCALLY FREE MODULE ON A PROPER CURVE SCHEME EQUALS THE DEGREE OF ITS DETERMINANT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Existence of a modification making a locally free module filtrable by invertible quotients. Stacks Project, Divisors, Lemma divisors-lemma-filter-after-modification (tag 0AYP). Produces the filtration on which the determinant computation is performed.
- Additivity and multiplicativity of the degree, and the determinant of a short exact sequence. Stacks Project, Varieties, Lemmas lemma-degree-additive (tag 0AYS) and lemma-degree-tensor-product (tag 0AYX); Modules on Sites, Lemma modules-lemma-det-ses (tag 0B38). Supply the inductive determinant identity and the final degree computation.
- Birational invariance of the degree and the componentwise description. Stacks Project, Varieties, Lemma lemma-degree-birational-pullback (tag 0AYU) and Lemma lemma-degree-in-terms-of-components (tag 0AYW). Allow replacing X by the modification and reducing to a single proper curve.

Source reference. Original lemma 0AYW: sources/stacks-chow/tags.txt, source lines 13804–13804. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0AYP: sources/stacks-chow/tags.txt, source lines 13797–13797. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYU: sources/stacks-chow/tags.txt, source lines 13802–13802. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B38: sources/stacks-chow/tags.txt, source lines 13956–13956. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYX: sources/stacks-chow/tags.txt, source lines 13805–13805. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0AYS: sources/stacks-chow/tags.txt, source lines 13800–13800. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field, let X be a proper scheme of dimension ≤ 1 over k, and let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of rank n. Then*

$$\deg(\mathcal{E}) = \deg(\wedge^n(\mathcal{E})) = \deg(\det(\mathcal{E}))$$

Proof. By Lemma 0AYW and elementary arithmetic, we reduce to the case of a proper curve. Then there exists a modification $f : X' \to X$ such that $f^*\mathcal{E}$ has a filtration whose successive quotients are invertible modules, see Divisors, Lemma 0AYP. By Lemma 0AYU we may work on X'. Thus we may assume we have a filtration

$$0 = \mathcal{E}_0 \subset \mathcal{E}_1 \subset \mathcal{E}_2 \subset \ldots \subset \mathcal{E}_n = \mathcal{E}$$

by locally free $\mathcal{O}_X$-modules with $\mathcal{L}_i = \mathcal{E}_i/\mathcal{E}_{i-1}$ is invertible. By Modules, Lemma 0B38 and induction we find $\det(\mathcal{E}) = \mathcal{L}_1 \otimes \ldots \otimes \mathcal{L}_n$. Thus the equality follows from Lemma 0AYX and additivity (Lemma 0AYS). □